

Assignment 2. Mujaab Badalo. 285269.

Synchronous generator with:

$$U_n = 10,5 \text{ kV}$$

$$S_n = 62,5 \text{ MVA}$$

$$\cos \phi_n = 0,8$$

$$x_d = 1,94 \%$$

$$U_s = 118 \text{ kV}$$

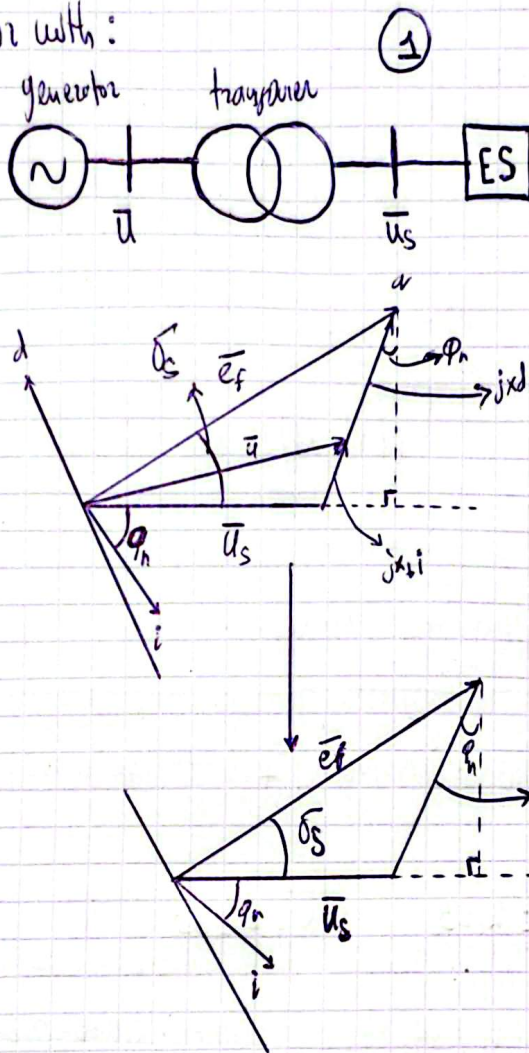
$$S_{nT} = 65 \text{ MVA}$$

$$P_T = 121 / 10,5$$

$$u_k = 10\%$$

$$P = 50 \text{ MW}$$

$$Q = 32 \text{ MVA}_2$$



$$\left[\begin{aligned} \bar{E}_f &= \bar{U}_s + j(x_t + x_d) \\ \text{Relative reactance: } x_t &= u_k \cdot \frac{U_{Tn}^2}{S_{Tn}} \cdot \frac{S_n}{U_n^2} \quad U_{Tn} = U_n \quad x_t = 0,096 \end{aligned} \right]$$

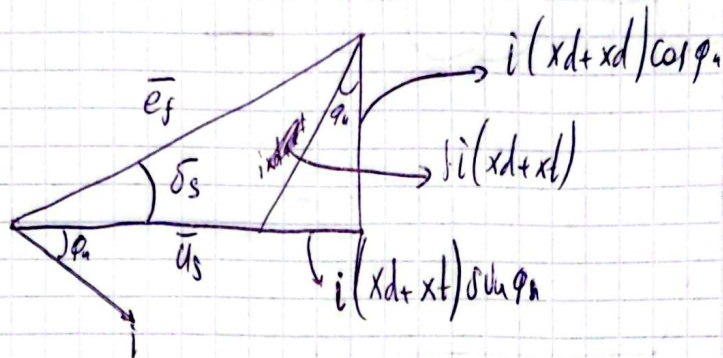
$$\left[\begin{aligned} \text{Synchronous reactance } x_d &= 1,94 \\ \text{Total reactance: } x_t + x_d &= 2,036 \end{aligned} \right]$$

$$\left[\text{Relative power grid voltage: } u_s = \frac{U_s}{U_n} \cdot \frac{1}{P_T} = \frac{118}{10,5} \cdot \frac{10,5}{121} = 0,975 \right]$$

$$\left[\begin{aligned} i &= \frac{I}{I_w} = \frac{\sqrt{P^2 + Q^2}}{\sqrt{3} \cdot U_s \cdot \frac{1}{P_T}} \cdot \frac{U_n \sqrt{3}}{S_n} = \\ &= \frac{\sqrt{50^2 + 32^2} \cdot 10,5 \sqrt{3} \cdot 121}{\sqrt{3} \cdot 118 \cdot 62,5 \cdot 10,5} = \frac{59,36 \cdot 121}{118 \cdot 62,5} = 0,974 \end{aligned} \right]$$

$$\left[\cos \varphi_n = \frac{P}{S} = \frac{P}{\sqrt{P^2 + Q^2}} = 0,84 \right]$$

$$\left[\sin \varphi_n = \frac{Q}{\sqrt{P^2 + Q^2}} = 0,54 \right]$$



$$\bar{E}_f = \sqrt{(U_s + i(x_d + x_t) \sin \varphi_n)^2 + (i(x_d + x_t) \cos \varphi_n)^2} =$$

$$= \sqrt{(0,975 + 0,974(1,94 + 0,096) \cdot 0,54)^2 + (0,974(1,94 + 0,096) \cdot 0,84)^2} =$$

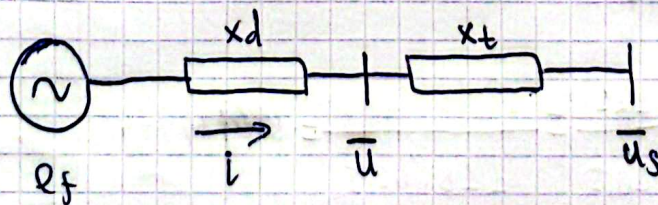
$$= \sqrt{4,185 + 2,775} = 2,63 \quad \bar{E}_f = 2,63$$

$$\arctan \delta_s = \frac{i(x_d + x_t) \cos \varphi_n}{U_s + i(x_d + x_t) \sin \varphi_n} = \frac{0,974 \cdot 2,036 \cdot 0,84}{0,975 + 0,974 \cdot 2,036 \cdot 0,54} =$$

$$\arctan \delta_s = \frac{1,66}{2,045} = 0,809 \quad \left[\delta_s = 38,99^\circ \right]$$

$$\left[E_f = \frac{e_f \cdot U}{\sqrt{3}} = \frac{2,63 \cdot 10,5}{\sqrt{3}} = 15,94 \text{ kV} \right]$$

$$\left[E_f = 15,94 \text{ kV} \quad \delta_s = 38,99^\circ \right]$$



② Active power dependence on torque angle :

$$\left[\begin{aligned} P(\delta_s) &= \frac{e_s \cdot u_s}{x_d + x_t} \cdot \sin \delta_s \\ Q(\delta_s) &= \frac{e_s \cdot u_s}{x_d + x_t} \cdot \cos \delta_s - \frac{u_s^2}{x_d + x_t} \end{aligned} \right]$$

When torque angle $\delta_s = 90^\circ$ Active power will be max:

$$P(90^\circ) = \frac{e_s \cdot u_s}{x_d + x_t} = \frac{2,63 \cdot 0,975}{2,036} = 1,26$$

$$\left[P(\text{at stability limit}) = 1,26 \cdot 62,5 = 78,75 \text{ MW} \right]$$

$$\text{Operating Power} = 50 \text{ MW}$$

$$\left[\text{Static stability reserve} : 78,75 - 50 = 28,75 \text{ MW} \right]$$

Power-Angle Curve for Turboalternator

